```
import numpy as np
import pandas as pd
import sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('advertising.csv')
df.head(9)
```

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9
5	8.7	48.9	75.0	7.2
6	57.5	32.8	23.5	11.8
7	120.2	19.6	11.6	13.2
8	8.6	2.1	1.0	4.8

```
df.shape
```

```
(200, 4)
```

```
df.describe()
```

	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	15.130500
std	85.854236	14.846809	21.778621	5.283892
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	11.000000
50%	149.750000	22.900000	25.750000	16.000000
75%	218.825000	36.525000	45.100000	19.050000
max	296.400000	49.600000	114.000000	27.000000

```python
df.tail(9)
```

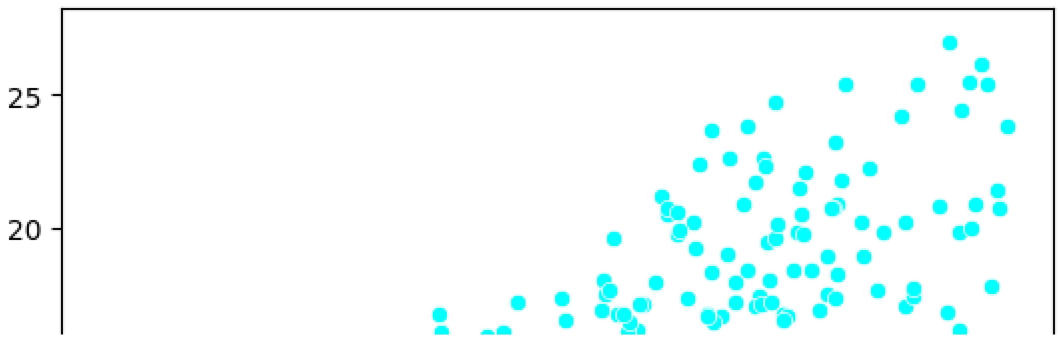
	TV	Radio	Newspaper	Sales
191	75.5	10.8	6.0	11.9
192	17.2	4.1	31.6	5.9
193	166.8	42.0	3.6	19.6
194	149.7	35.6	6.0	17.3
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	14.0
197	177.0	9.3	6.4	14.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4

```python
df.isnull().sum()
```

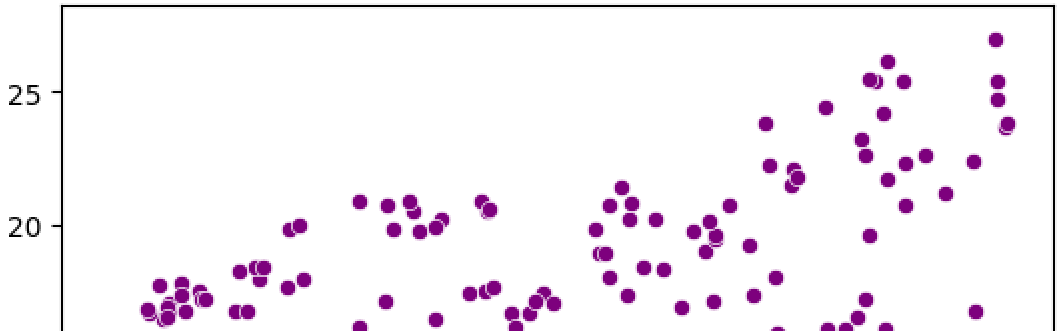
	0
TV	0
Radio	0
Newspaper	0
Sales	0

dtype: int64

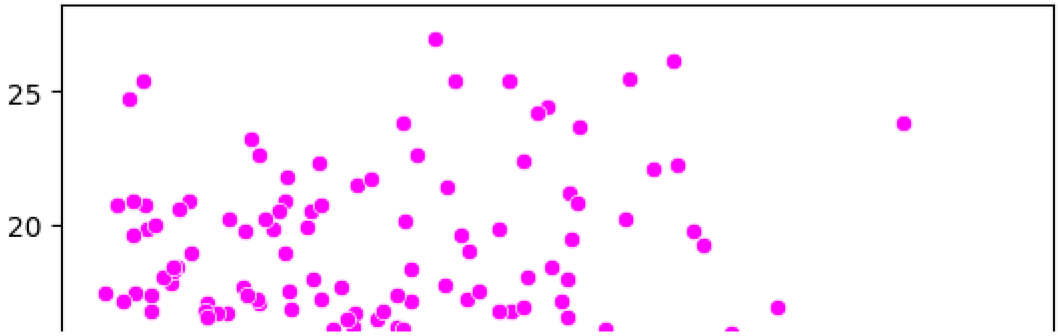
```python
sns.scatterplot(x=df['TV'], y=df['Sales'], color='cyan')
plt.show()
```



```
sns.scatterplot(x=df['Radio'], y=df['Sales'], color='purple')
plt.show()
```

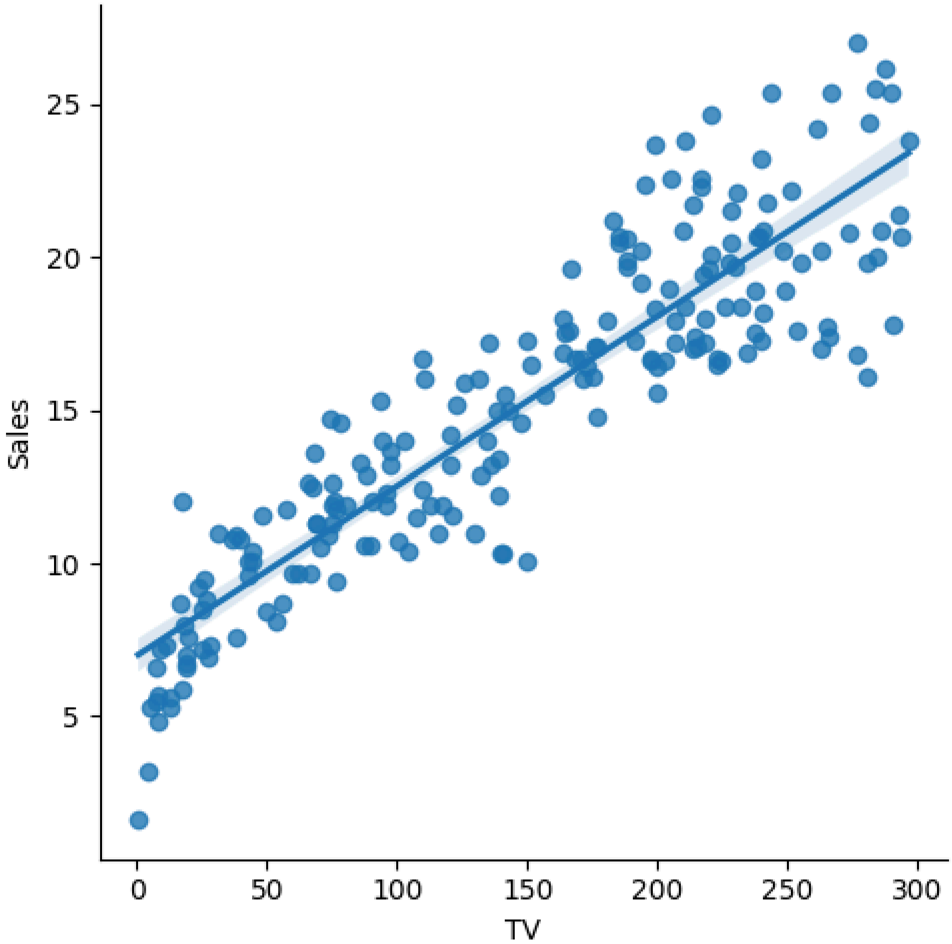


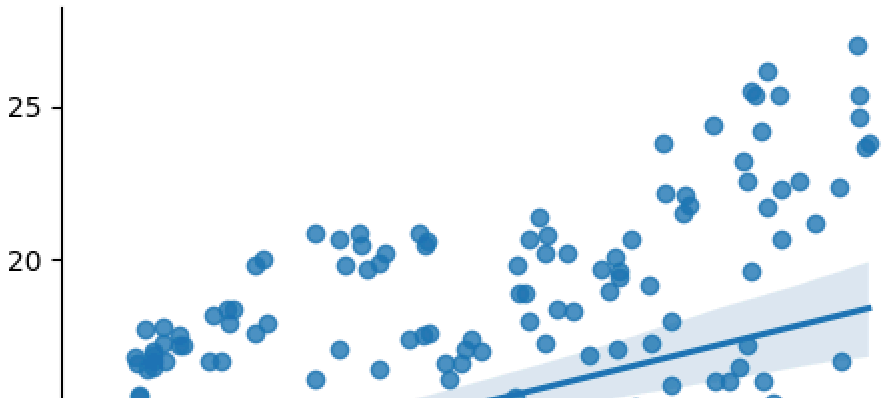

```
sns.scatterplot(x=df['Newspaper'], y=df['Sales'], color='magenta')
plt.show()
```



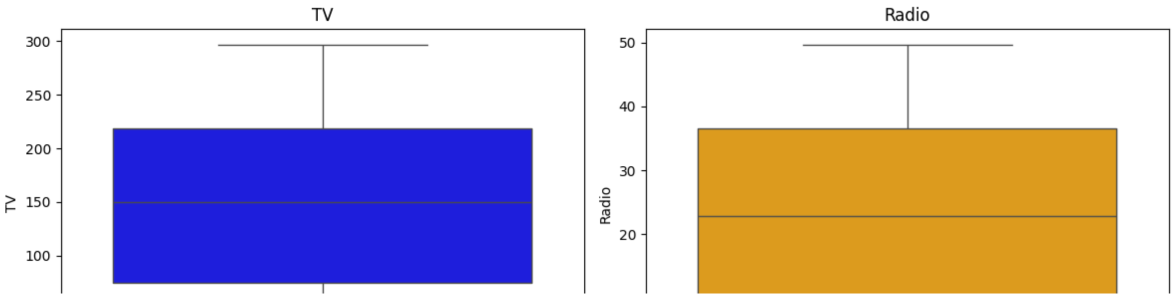
```
sns.lmplot(x='TV', y='Sales', data=df)
sns.lmplot(x='Radio', y='Sales', data=df)
sns.lmplot(x='Newspaper', y='Sales', data=df)
```

<seaborn.axisgrid.FacetGrid at 0x7ba0e2918910>

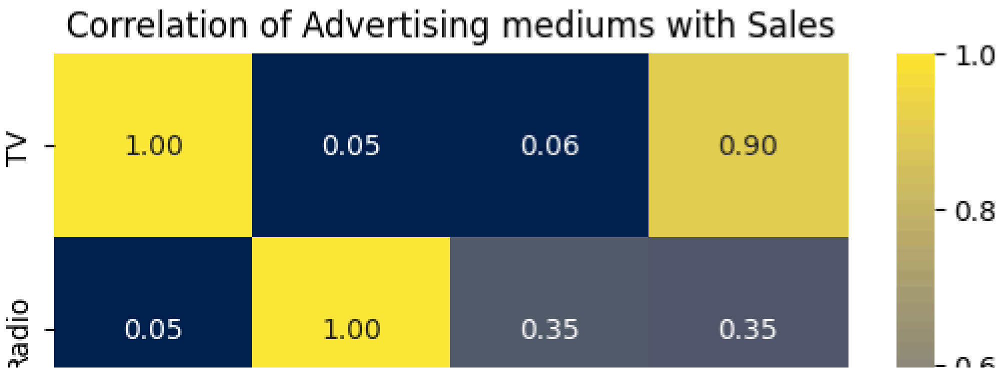




```
plt.figure(figsize=(12, 8))

plt.subplot(221)
sns.boxplot(y=df['TV'], color='Blue')
plt.title('TV')

plt.subplot(222)
sns.boxplot(y=df['Radio'], color='Orange')
plt.title('Radio')

plt.subplot(223)
sns.boxplot(y=df['Newspaper'], color='Green')
plt.title('Newspaper')

plt.subplot(224)
sns.boxplot(y=df['Sales'], color='Red')
plt.title('Sales')

plt.tight_layout()
plt.show()
```



```
sns.heatmap(df.corr(), annot=True, fmt='.2f', cmap='cividis')
plt.title('Correlation of Advertising mediums with Sales')
plt.show()
```




```
X = df[['TV']]
Y = df[['Sales']]
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, rar
```

```
print(X_train,X_test,Y_train,Y_test)
```

```
[[13.44312992]
 [16.8275895 ]
 [ 9.39842331]
 [10.46369584]
```

```
 [19.43528787]
 [ 9.12655689]
 [10.92420427]
 [15.17974607]
 [12.81062436]
 [11.24045705]
 [11.34587464]
 [16.35043619]
 [ 7.48981004]
 [ 7.43987539]
 [11.24600534]
 [14.18660137]
 [11.07955651]
 [23.08051729]
 [ 8.09457413]
 [17.97053815]
 [22.78090939]
 [17.26590476]
 [13.2711328 ]
 [ 8.32760249]
 [23.13600023]
 [ 8.08347754]
 [23.29690077]
 [ 8.04463948]
 [14.4584678 ]
 [ 8.42747179]
 [12.577596  ]
 [11.45684053]
 [17.46009506]
 [16.8275895 ]
 [13.98131448]
 [18.63633348]
 [14.93562112]
 [17.26590476]
 [19.34651516]
 [20.41733598]
 [ 7.96141506]
 [13.69280317]
 [11.9839285 ]
 [17.60989901]
 [11.1960707 ]
 [17.7264132 ]
 [11.76199672]
 [21.81550616]
 [ 9.19868471]
 [21.50480168]
 [ 7.73948329]
 [17.75415467]
 [23.45225301]
 [21.74337834]
 [18.91929649]
 [15.31290513]
 [14.31421214]
```

```
df_2 = LinearRegression()
df_2.fit(X_train, Y_train)
```

```
▾ LinearRegression

LinearRegression()
```

```
X_train = df_2.predict(X_train)
print(X_train)
```

```
[[13.44312992]
 [16.8275895 ]
 [ 9.39842331]
 [10.46369584]
 [19.43528787]
 [ 9.12655689]
 [10.92420427]
 [15.17974607]
 [12.81062436]
 [11.24045705]
 [11.34587464]
 [16.35043619]
 [ 7.48981004]
 [ 7.43987539]
 [11.24600534]
 [14.18660137]
 [11.07955651]
 [23.08051729]
 [ 8.09457413]
 [17.97053815]
 [22.78090939]
 [17.26590476]
 [13.2711328 ]
 [ 8.32760249]
 [23.13600023]
 [ 8.08347754]
 [23.29690077]
 [ 8.04463948]
 [14.4584678 ]
 [ 8.42747179]
 [12.577596  ]
 [11.45684053]
 [17.46009506]
 [16.8275895 ]
 [13.98131448]
 [18.63633348]
 [14.93562112]
 [17.26590476]
 [19.34651516]
 [20.41733598]
 [ 7.96141506]
 [13.69280317]
 [11.9839285 ]
 [17.60989901]
 [11.1960707 ]
 [17.7264132 ]
 [11.76199672]
 [21.81550616]
 [ 9.19868471]
 [21.50480168]
 [ 7.73948329]
```

```
 [17.75415467]
 [23.45225301]
 [21.74337834]
 [18.91929649]
 [15.31290513]
 [14.31421214]
 [10.1973777 ]
```

```
X_test = df_2.predict(X_test)
print(X_test)
```

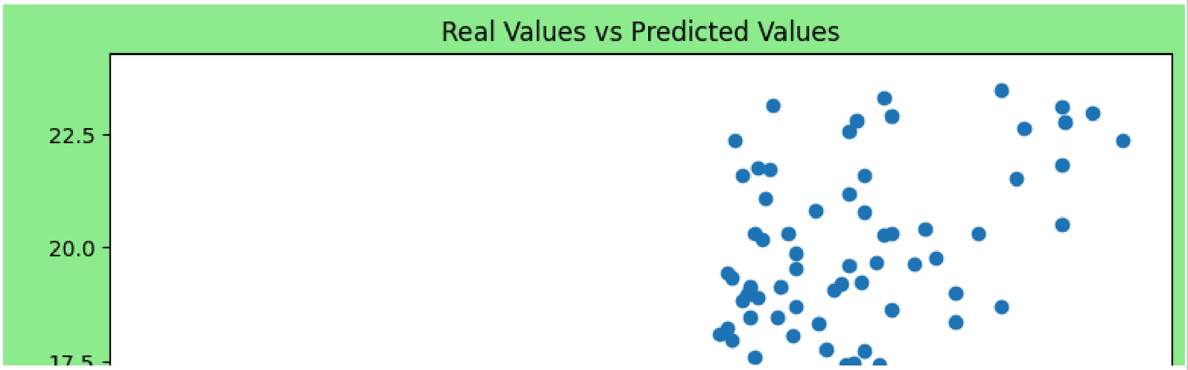
```
[[16.06747317]
 [17.84847567]
 [23.25806271]
 [ 7.65625887]
 [19.23000098]
 [11.17387752]
 [19.03581067]
 [ 9.78125562]
 [19.34651516]
 [16.72217191]
 [ 8.75482116]
 [10.12524988]
 [20.01785878]
 [ 7.30671633]
 [14.74697911]
 [16.45030549]
 [ 7.41213392]
 [17.97053815]
 [11.18497411]
 [20.17875932]
 [19.74044406]
 [10.76885203]
 [ 9.1154603 ]
 [20.92777906]
 [10.83543156]
 [ 9.97544593]
 [18.85271696]
 [14.73588252]
 [11.84522114]
 [ 7.47316516]
 [18.09260063]
 [10.84652815]
 [18.04266598]
 [ 7.94477018]
 [22.58117079]
 [20.22314567]
 [ 9.68693462]
 [22.19279018]
 [13.50970946]
 [ 8.53288939]]
```

```
print(Y_test)
```

```
     Sales
95    16.9
15    22.4
30    21.4
158    7.3
```

```
128    24.7
115    12.6
69     22.3
170     8.4
174    16.5
45     16.1
66     11.0
182     8.7
165    16.9
78      5.3
186    10.3
177    16.7
56      5.5
152    16.6
82     11.3
68     18.9
124    19.7
16     12.5
148    10.9
93     22.2
65     11.3
60      8.1
84     21.7
67     13.4
125    10.6
132     5.7
9      15.6
18     11.3
55     23.7
75      8.7
150    16.1
104    20.7
135    11.6
137    20.8
164    11.9
76      6.9
```

```
plt.figure(figsize=(9, 7), facecolor='lightgreen', edgecolor='red', alpha=1)
plt.scatter(Y_train, X_train)
plt.xlabel('Real Values')
plt.ylabel('Predicted Values')
plt.title('Real Values vs Predicted Values')
plt.show()
```



```
plt.figure(figsize=(9, 7), facecolor='lightgreen', edgecolor='red', alpha=1)
plt.scatter(Y_test, X_test)
plt.xlabel('Real Values')
plt.ylabel('Predicted Values')
plt.title('Real Values vs Predicted Values')
plt.show()
```